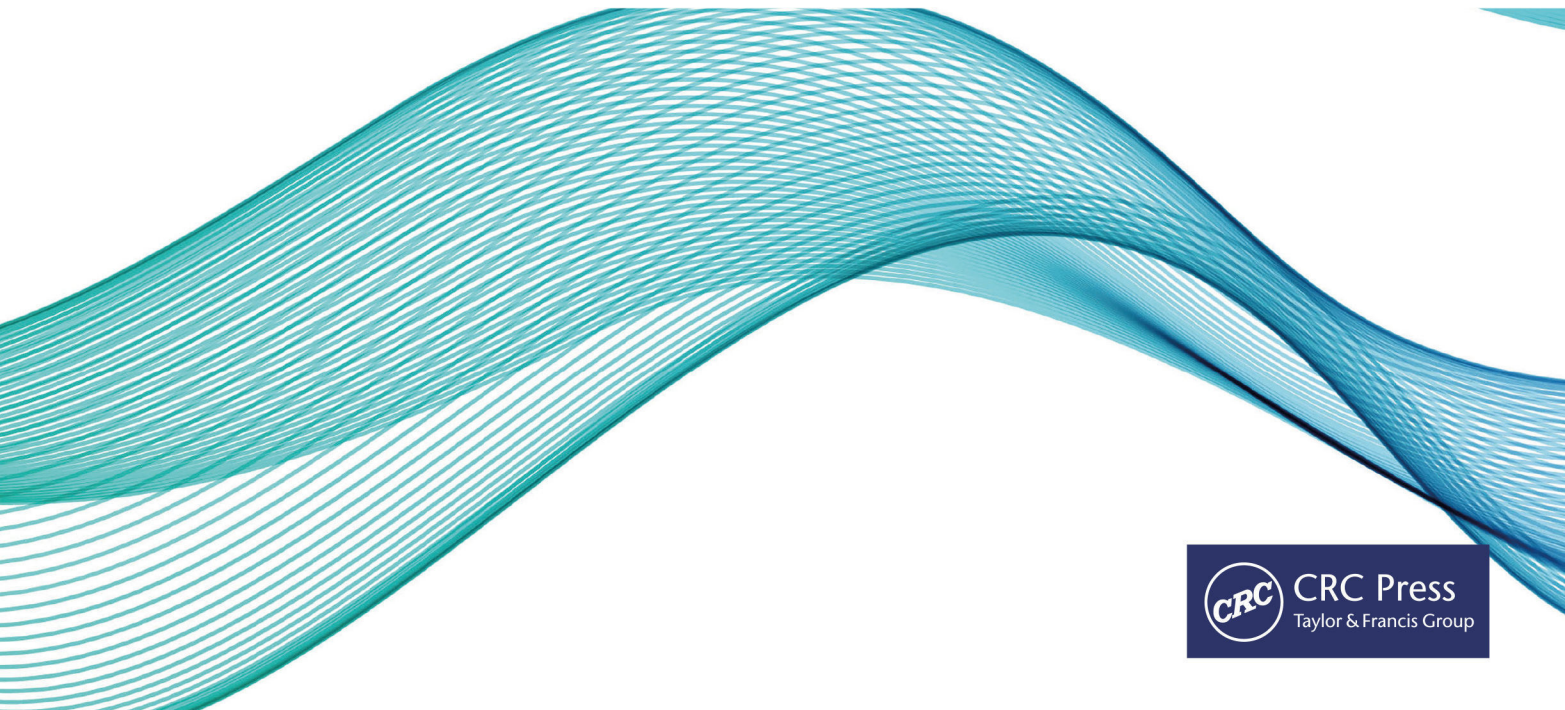


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Debashis De, Panagiotis Sarigiannidis, Jan Platoš
and Muhammad Mujtaba Asad



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Preface

ICSCCA is a premier conference on Smart Computing, Communication Technology & Automation that promotes an open forum where researchers, scientists, and engineers from academia and industries come together to present current and exciting research applications and discuss future challenges in these fascinating fields. It aims to generate activities in research, development, and application in Smart Computing, Communication Technology & Automation by exchanging knowledge and experiences & fostering synergy among research groups from many nations.

The conference received an overwhelming response, attracting more than 251 research paper submissions from across the globe. After a rigorous review process, 45 papers were accepted and registered for presentation at the conference. By fostering interdisciplinary collaboration and showcasing impactful innovations, ICSCCA 2025 aims to inspire sustainable innovation, technological growth, and broader societal benefits. For years to come, may this compilation of proceedings act as a lighthouse of understanding, encouraging fresh discoveries, creative thinking, and insightful conversations.

The First International Conference on Smart Computing, Communication & Automation (ICSCCA 2025), organized by the Budge Budge Institute of Technology, Kolkata, India, at its campus from August 2nd and 3rd, 2025, in collaboration with Kyongdong University, South Korea. We would like to express our sincere gratitude to Mr. Jagannath Gupta, the Honorable Chairman, BBIT, for his visionary leadership and unwavering support, which have been crucial to the success of this conference.

The ICSCCA 2025 conference would like to express its appreciation for the outstanding contributions made by the Scientific, Advisory, and Technical Program Committee members, as well as the Keynote speakers who delivered their presentations during 2nd and 3rd August, 2025. We also want to officially thank the reviewers for their excellent contributions in terms of their time, effort, and devotion. We also like to thank the session chairs for their support in ensuring the sessions ran smoothly.

We also like to thank everyone who attended the conference, including the speakers and participants. We really appreciate everyone who contributed to and assisted with this conference. We would like to extend our sincere gratitude and appreciation for all the hard work and dedication provided by the members of the organizing committee, program committee, volunteers, and staff serving ICSCCA 2025. We also wish to thank our Chief Patron, Patron, and the members of various conference committees for their support.

Finally, we would like to express our gratitude to Mrs. Shatakshi Mishra, the Editorial Manager at CRC Press, and the entire CRC Press editorial team for their kind support in publishing this book.

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Dedication

This compilation of conference proceedings exemplifies a shared vision, forged by curiosity, creativity, and steadfast commitment to the quest for knowledge. The proceedings of ICSCCA-2025 are earnestly dedicated to all researchers, practitioners, reviewers, academicians, and visionaries whose relentless efforts and intellectual contributions have consistently advanced the domains of smart computing, communication, and automation, along with their diverse applications, to new heights. Their dedication to innovation and academic achievement has enhanced this meeting and strengthened the wider research community.



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About the Editors

Himadri Biswas currently works as a Dean and Professor in the department of Computer Science and Engineering and immediate past HoD and Professor in the Department of Information Technology of Budge Budge Institute of Technology, Kolkata. He has 22 years of teaching experience and served at several institutions as a HoD and Professor in the Department of Computer Science and Engineering of Dr. Sudhir Chandra Sur Institute of Technology and Sports Complex under JIS Group, Assistant Professor in the Department of Computer Science & Engineering of Bengal College of Engineering & Technology, Durgapur and Modern Institute of Engineering & Technology, Bandel, Hooghly; before that he worked as a lecturer in the Dept. of Computer Applications of Techno India Hooghly, West Bengal, India. He completed his Ph.D. degree in Computer Science and Engineering from the University of Kalyani, Nadia, West Bengal, India. He completed his M. Tech in Computer Science & Engineering from WBUT and his MCA from St. Xavier's College in Kolkata under Indira Gandhi National University, New Delhi, India. He has the owner of different National and International patents, a number of research articles published in different national and international journals indexed with SCI & SCOPUS, conference papers and book chapters. His research interests include cloud computing, mobile computing, mobile cloud computing, data science, machine learning and deep learning.

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Siddhartha Bhattacharyya (FIET(UK), FIEI, FIETE, LFOSI, FSCRS, SMIEEE, SMACM, SMAAIA, SMIETI, LMCSI, LMISTE) is currently serving as a Senior Researcher in the Faculty of Electrical Engineering and Computer Science of VSB Technical University of Ostrava, Czech Republic. He also serves as the Scientific Advisor of Algebra University College, Zagreb, Croatia. Before this, he served as the Principal of Rajnagar Mahavidyalaya, Rajnagar, Birbhum. He served as a Professor in the Department of Computer Science and Engineering of Christ University, Bangalore. He was the Principal of RCC Institute of Information Technology, Kolkata, India. He has also served as a Senior Research Scientist in the Faculty of Electrical Engineering and Computer Science of VSB Technical University of Ostrava, Czech Republic. He is the recipient of several coveted awards, including the Distinguished HoD Award and Distinguished Professor Award conferred by the Computer Society of India, Mumbai Chapter, India, in 2017, the Honorary Doctorate Award (D. Litt.) from The University of South America, and the South East Asian Regional Computing Confederation (SEARCC) International Digital Award ICT Educator of the Year in 2017. He was an ACM Distinguished Speaker for 2018–2020. He was inducted into the People of ACM Hall of Fame by ACM, USA, in 2020. He was the IEEE Computer Society Distinguished Visitor for 2021–2023. He is a full foreign member of the Russian Academy of Natural Sciences (RANS) and the Russian Academy of Engineering (REA). He is a full fellow of The Royal Society for the Arts, Manufacturers and Commerce (RSA), London, UK. He is a co-author of seven books, the co-editor of 114 books and has more than 400 research publications in international journals and conference proceedings to his credit.

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Panagiotis Sarigiannidis is the Director of the ITHACA lab, co-founder of the 1st spin-off of the University of Western Macedonia: MetaMind Innovations P.C., and Full Professor in the Department of Electrical and Computer Engineering in the University of Western Macedonia, Kozani, Greece. He received B.Sc. and Ph.D. degrees in computer science from the Aristotle University of Thessaloniki, Thessaloniki, Greece, in 2001 and 2007, respectively. He has published over 360 papers in international journals, conferences and book chapters. He received six best paper awards and the IEEE SMC TCHS Research and Innovation Award 2023. He has been involved in several national, European and international projects, coordinating and technically leading numerous national and European projects including H2020, Horizon Europe, Erasmus+ and operational programs. His research interests include telecommunication networks, internet of things and network security. He is an IEEE member and participates in the Editorial Boards of various journals.

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Enhancing Interpretability in Chronic Kidney Disease Prediction: A Comparative Analysis of SHAP, LIME and Permutation Feature Importance

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ABSTRACT: Chronic kidney disease (CKD) is a progressive condition often caused by diabetes and hypertension. The global morbidity and mortality rate of CKD pose a significant burden on the healthcare system. Machine learning (ML) models have shown promise for early CKD prediction, but their "black box" nature limits clinical adoption. This article evaluates multiple ML models for CKD prediction, highlighting XGBoost as the best performer. To enhance transparency, the study integrates Explainable AI (XAI) methods, SHAP, LIME, and Permutation Feature Importance, providing insights into model behavior. Serum Creatinine and GFR are identified as key predictors irrespective of the ML models. By comparing the strengths and weaknesses of XAI techniques, this research promotes trust in ML-driven CKD predictions. It is offering a framework for ethical AI integration into clinical practice for CKD management.

Keywords: Chronic Kidney Disease (CKD) prediction, Explainable Artificial Intelligence (XAI) for CKD, Local Interpretable Model-agnostic Explanations (LIME) for CKD, Comparing XAI for CKD

1 INTRODUCTION

Chronic kidney disease (CKD) is advancing condition identified by damaged kidney or reduced function, typically assessed by the Glomerular Filtration Rate (GFR). A GFR lower than 60 mL/min per 1.73 m^2 for more than three months, or the presence of other kidney damage markers generally indicates CKD [1]. Common causes include diabetes, hypertension, and various other conditions [2]. Due to its often-asymptomatic progression, patients frequently present with complications such as anemia, cardiovascular issues, or neurological problems by the time they seek medical care. The global burden of CKD is escalating rapidly, with significant increases in incidence, prevalence, and mortality for past few decades [3,4].

Machine learning (ML) models have demonstrated considerable promise in the early detection and diagnosis of CKD [5,6] and other diseases [7–10]. However, many of these models are regarded as "black boxes", making the process of taking decisions difficult to understand for clinicians. This absence of transparency hinders their integration into clinical practice [11]. Furthermore, the complexity of ML-based diagnostic systems poses challenges for nephrologists, particularly in low-resource settings where specialized medical professionals are often scarce.

To address these challenges, Explainable Artificial Intelligence (XAI) methods, including SHAP (SHapley Additive exPlanations), LIME (Local Interpretable Model-agnostic Explanations), and permutation feature importance, have been developed [13–15]. These techniques provide deep understanding of the decision-making processes of ML models by categorizing contributions of each attributes to the predictions. By integrating these explainability techniques into CKD prediction models, clinicians can better understand the underlying factors influencing predictions, ultimately enhancing trust in automated systems

Comparing different XAI methods, such as SHAP, LIME, and permutation feature importance, is valuable because it offers a comprehensive understanding of how features contribute to model predictions by leveraging each method's unique perspective. This comparison reveals the strengths and limitations of each technique—such as SHAP's global consistency versus LIME's focus on local interpretability—helping in selecting the most suitable method for specific use cases. Additionally, comparing these techniques can guide model improvements by highlighting the most impactful features, refining feature selection and model tuning processes. It also helps identify potential biases in predictions, ensuring fairness in outcomes, which is particularly important in healthcare. Finally, these comparative insights can inform policy and regulatory frameworks by guiding the use of appropriate XAI techniques in sensitive areas like medical diagnostics, promoting ethical and transparent AI use.

This study emphasizes the joint application of SHAP, LIME, and permutation feature importance on a CKD dataset to bolster the explainability of CKD pre dictions. By leveraging these methods in tandem, we aim to provide a comprehensive understanding of model behavior and promote transparency in the decision-making process. Our research adds up to the expanding field of explainable AI (XAI) in healthcare, improving the trustworthiness of ML models for CKD diagnosis and facilitating the ethical adoption of AI-enabled tools in clinical usage. This paper makes the following key contributions:

1. **Evaluation of ML Classifiers for CKD Prediction:** The study thoroughly assessed multiple machine learning models, which includes Gradient Boosting Machines (GBM), XGBoost, and CatBoost, identifying XGBoost as the best-performing classifier with an accuracy of 93.49%. This performance analysis contributes to the selection of robust models suitable for CKD diagnosis.
2. **Integration of Explainable AI (XAI) Techniques:** To handle the "black-box" issue of ML models, the study used SHAP, LIME, and Permutation Feature Importance, offering both global and local insights for more transparent CKD predictions.
3. **dentification of Critical CKD Predictors:** Across all XAI methods, Serum Creatinine and Glomerular Filtration Rate (GFR) were consistently recognized as the most influential features. This finding strengthens the clinical validity of these biomarkers while highlighting the model's ability to integrate both symptoms and biomarkers.
4. **Comparison of XAI Methods:** By evaluating the strengths and limitations of SHAP, LIME, and Permutation Feature Importance, this paper contributes to understanding how these methods complement each other, with SHAP excelling in capturing feature interactions and LIME in pro viding

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instance-specific explanations. This comparative analysis guides clinicians and researchers in selecting the most appropriate XAI techniques for different healthcare applications.

5. **Enhanced Clinical Decision-Making:** The study enhances trust in ML driven CKD predictions by making the models' decision-making processes more interpretable for clinicians. The improved transparency fosters ethical AI integration into clinical practice, facilitating better-informed decisions in CKD diagnosis and management. By contributing to both the development of accurate CKD models and their interpretability, this work advances the field of AI in healthcare, promoting trust and the responsible adoption of machine learning for clinical decision support.

2 METHODOLOGY

2.1 *Data collection*

The Chronic Kidney Disease (CKD) dataset used in this study is obtained from Kaggle (https://www.kaggle.com/datasets/rabieelkharoua/chronic-kidney-disea se-dataset-analysis) and contains 1,659 patient records with 54 features (Table 1). This dataset provides a comprehensive collection of demographics, clinical, and biochemical attributes related to CKD, which makes it suitable for machine learning-based analysis aimed at diagnosing kidney disease. Each patient entry includes both categorical and numerical variables that describe various aspects of their health and lifestyle. Interestingly, the dataset is clean and free from null values. The richness of the dataset provides an ideal foundation for evaluating machine learning models aimed at diagnosing CKD, with multiple features reflecting the complex, multifactorial nature of kidney disease. This dataset is utilized to assess explainability and feature importance through various eXplainable Artificial Intelligence (XAI) techniques, including SHAP, LIME and Permutation Feature Importance.

Table 1. Categories and key features of CKD dataset.

Category	Key Features
Demographic Information	PatientID, Age, Gender, Ethnicity, Socioeconomic Status, EducationLevel
Lifestyle Factors	BMI, Smoking, Alcohol Consumption, Physical Activity, Diet Quality, Sleep Quality
Medical and Family History	Family History, Kidney Disease, Family History, Hypertension, Family History, Diabetes, Previous Acute Kidney Injury, Urinary Tract Infections
Clinical Measurements	Systolic BP, Diastolic BP, Fasting Blood Sugar, HbA1c, Serum Creatinine, BUN Levels, GFR, Protein in Urine, ACR, Serum Electrolytes (Sodium, Potassium, Calcium, Phosphorus)
Cholesterol and Medications	Cholesterol Total, Cholesterol LDL, Cholesterol HDL, Cholesterol Triglycerides, ACE Inhibitors, Diuretics, NSAIDsUse, Statins, Antidiabetic Medications
Symptomatology	Edema, Fatigue Levels, Nausea Vomiting, Muscle-Cramps, Itching
Additional Health Indicators	Quality of LifeScore, Heavy Metals Exposure, Occupational Exposure Chemicals, Water Quality, Medical Checkups Frequency, Medication Adherence, Health Literacy
Outcome Variable	Diagnosis (CKD or non-CKD)

2.2 *Machine learning algorithms*

This study employs a comprehensive set of machine learning classifiers to analyze Chronic Kidney Disease (CKD) prediction. The classifiers included Logistic Regression, Decision Tree (DT), Random Forest (RF), Extra Trees (ET), Support Vector Machine (SVM), k-Nearest Neighbors (k-NN), Naive Bayes (Gaussian Naive Bayes, GNB), Gradient Boosting Machines (GBM), AdaBoost, XGBoost, LightGBM, CatBoost, Multilayer Perceptron (MLP), Linear Discriminant Analysis (LDA), Quadratic Discriminant Analysis (QDA), and Stochastic Gradient Descent (SGD) Classifier.

Each of these classifiers offers unique advantages for model development. For instance, Decision Tree generates a tree-like pattern by recursively dividing the data to minimize impurity, while Random Forest boosts accuracy by considering predictions from multiple decision trees. Extra Trees introduces further randomization in tree construction to reduce variance and computational cost. SVM constructs a hyper-plane to maximize the margin between classes, in contrast, GNB assumes feature independence and uses the Bayes theorem.

k-NN, a distance-based algorithm, classifies data based on majority voting among neighboring points. Boosting techniques like GBM, AdaBoost, XGBoost, LightGBM, and CatBoost iteratively enhance model performance by focusing on correcting previous errors, particularly useful for weak learners. The neural network-based MLP adjusts weights through backpropagation to know complex patterns in data, while LDA and QDA aim to maximize class separation, with QDA allowing quadratic decision boundaries. Finally, SGD optimizes the model by updating parameters iteratively to minimize the classification error. This diverse selection of algorithms allowed us to evaluate different perspectives for CKD prediction, ranging from simple linear models to complex ensemble techniques.

2.3 *Explainable AI (XAI) Techniques for Model Interpretability*

To complement our machine learning models and ensure interpretability, especially in a critical domain like healthcare, we utilized various Explainable AI (XAI) techniques. These methods provided insights into feature importance and model behavior, which are crucial for building trust and transparency in predictive models. In this study, we implemented SHAP, LIME and Permutation Feature Importance to understand the predictions made by our models.

2.3.1 *SHAP (SHapley Additive exPlanations)*

In order to quantify each feature's contribution to a model's prediction in relation to a baseline, usually the average prediction, SHAP, which is based on cooperative game theory, provides each feature a Shapley value. The Shapley value $\phi_i$of a feature $i$is computed as:

$$\phi_i = \sum_{S \subseteq N \setminus \{i\}} \frac{|S|!(|N| - |S| - 1)!}{|N|!} [f(S \cup \{i\}) - f(S)]$$

where N is the set of features, S is a subset of features, and $f(S)$ represents the model's prediction for S. This approach provides both local and global explanations, making SHAP suitable for understanding individual predictions and overall model behavior across the dataset.

2.3.2 *LIME (Local Interpretable Model-agnostic Explanations)*

The process of approximating complex models using more easily comprehensible models in order to understand specific predictions is done through LIME, such as linear models, around a specific instance. LIME perturbs the input data to create synthetic samples and fits a simple model to these perturbed instances to highlight the most important features. The loss function minimized by LIME is:

$$\mathrm{L}(g, f, \pi_x) = \underset{z' \in Z}{X} \pi_x(z')(f(z') - g(z'))^2$$

where f is the original complex model, g is the interpretable local model, and π_x represents a proximity measure that weighs perturbed samples based on their similarity to the original instance.

2.4 *Permutation feature importance*

Permutation Feature Importance is used to measure the effect of randomly permuting feature values on model performance. The importance score I_j for feature j is calculated as:

$$I_j = \frac{1}{n}\sum_{i=1}^{n}(L(f(X), y) - L(f(\widetilde{X}_j), y))$$

where $L(f(X),y)$ represents the model's loss with the original feature values, $\widetilde{X}_j$ is the dataset with permuted feature j, and n is the number of instances. This method identifies features that, when disrupted, cause significant degradation in model performance, thus highlighting their importance.

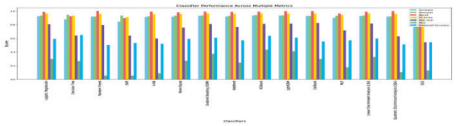
By incorporating XAI techniques such as SHAP, LIME and Permutation Feature Importance, we enhanced the interpretability of our CKD prediction models. These techniques not only provided a deeper understanding of the feature contributions but also ensured transparency, enabling us to verify that our models were making decisions based on medically relevant information. This is essential in healthcare, where the interpretability of predictive models directly affects clinical decision-making and patient outcomes.

3 RESULTS

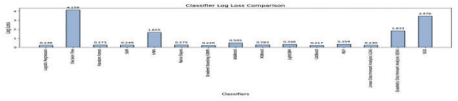
In this study, we conducted experiments in two phases. First, we thoroughly evaluated the competency of different machine learning classifiers to predict outcomes using the CKD dataset. Ten-fold cross-validation was used to evaluate these classifiers, taking accuracy into account, precision, AUC-ROC scores, and other key performance metrics to determine their effectiveness in predicting CKD. In the second phase, we focused on feature importance analysis through Explainable AI (XAI) techniques. We applied three XAI methods—SHAP, LIME and Permutation Feature Importance—to identify the most influential features driving CKD predictions for our proposed model. By comparing the important features derived from these XAI techniques, we obtained both global and local insights into the factors influencing the models' decisions. This comprehensive approach enabled the robust selection of impactful features, offering a deeper understanding of the critical predictors in CKD models.

3.1 *Assessing the performance of the classifiers*

In the evaluation of classifiers (Figure 1), Gradient Boosting Machines (GBM), XGBoost, and CatBoost consistently emerged as the top performers. GBM achieved an Accuracy of 93.31%, while XGBoost slightly outperformed with 93.49%, and CatBoost followed closely with 93.01%. All three models had high Precision (above 0.93), near-perfect Recall (above 0.99), and strong F1-Scores (above 0.96). These metrics, combined with high ROC-AUC scores (above 0.82), demonstrate their robust ability to handle both class discrimination and predictive accuracy. In contrast, Random Forest and SVM performed well in terms of Recall but showed limitations in Balanced Accuracy (0.50), suggesting potential struggles with class imbalance. Models like Decision Tree and Stochastic Gradient Descent (SGD) demonstrated lower performance across several metrics, especially MCC and Balanced Accuracy, indicating they are not as suitable for this task. Based on these results, XGBoost has been considered as the best classifier model for further study. For the next use of explainability methods like Permutation Feature Importance, SHAP, and LIME, we will concentrate exclusively on XGBoost, as it demonstrated the highest overall performance and robustness across all key metrics.



(a) Performance comparison of classifiers



(b) Log-loss comparison of classifiers

Figure 1. Classifier Performance on the CKD dataset using 10-fold cross validation with train: test as 80:20.

3.2 *Explainability analysis using SHAP, LIME and permutation feature importance*

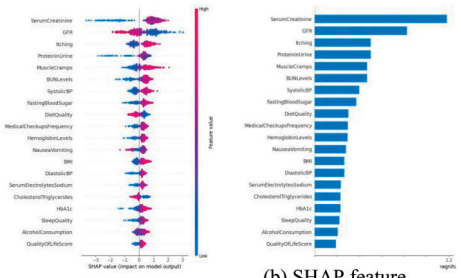
In this set of experiments the application of XAI techniques for CKD prediction is conducted. Particularly, we utilized SHAP (SHapley Additive exPlanations), LIME (Local Interpretable Model-agnostic Explanations), and Permutation Feature Importance to enhance the interpretability of predictions made by the XGBoost (XGB) classifier for Chronic Kidney Disease (CKD). These Explainable AI (XAI) tools were applied in both global and local contexts: SHAP summary plot, SHAP feature importance plot, and permutation feature importance for global explainability, and SHAP force plot alongside LIME for local interpretability.

3.2.1 *Comparing global feature importance techniques for predicting chronic kidney disease*

This section presents a comparative analysis of three global feature importance techniques: SHAP Summary Plot, SHAP Feature Importance, and Permutation Feature Importance. These processes help us to measure the impact of various features on the prediction given by the model. The SHAP summary plot (Figure 2(a)) provides a detailed explanation of how the feature values influences the prediction of the model. Features such as Serum Creatinine and GFR were the most significant, with higher values strongly influencing the CKD prediction, as indicated by the red color gradient. Symptom-related features, such as Itching and Muscle Cramps, also showed high SHAP values, highlighting their contribution to the model. Additionally, the SHAP plot revealed variations in feature behavior, such as how both high and low values of Protein in Urine, Fasting Blood Sugar, and BUN Levels contributed to the predictions, which were less apparent in other methods.

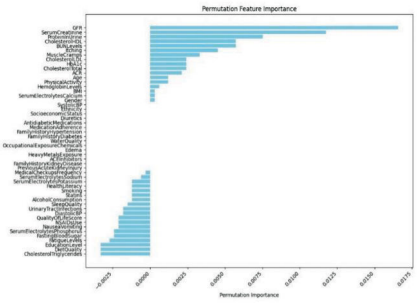
Features are ranked according to their average contribution to the model's output in the SHAP feature importance plot (Figure 2(b)). Serum Creatinine and GFR were the most significant features, much like the summary plot.Interestingly, symptoms like Itching and Muscle Cramps also ranked highly, which reinforces the model's ability to integrate subjective symptoms with objective biomarkers for CKD prediction. Although lifestyle factors such as Diet Quality and Sleep Quality ranked lower, their moderate contributions indicate some relevance to the model's predictions.

Permutation feature importance (Figure 2(c)) measures the decrease in model performance when each feature is shuffled. GFR and Serum Creatinine remained the top two features, underscoring their well-established clinical significance in



(a) SHAP summary plot for CKD

(b) SHAP feature importance plot for CKD dataset



(c) Permutation feature importance plot for CKD

Figure 2. Comparing global techniques for predicting chronic kidney disease.

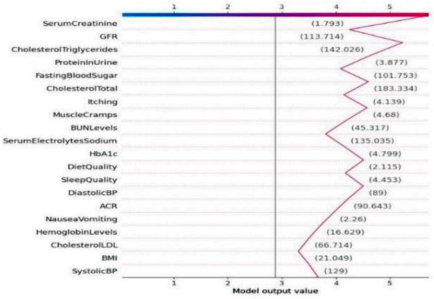


Figure 3. SHAP decision plot for CKD prediction.

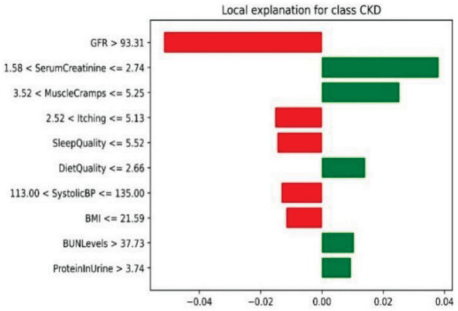


Figure 4. LIME plot for CKD prediction.

kidney function. Other biomarkers, including Protein in Urine, Cholesterol HDL, and BUN, also showed significant contributions. However, features such as Health Literacy, Diet Quality, and Cholesterol Triglycerides were ranked lower, suggesting minimal influence on the model's predictive performance.

Both the SHAP Feature Importance and Permutation Importance methods consistently ranked Serum Creatinine and GFR as the most critical predictors, validating the model's reliance on clinical biomarkers. Additionally, the SHAP Summary Plot provided deeper insights into the relationship between individual feature values and model output, revealing that both high and low feature values significantly influence predictions. A major limitation of the Permutation Importance method is its inability to account for feature interactions and the varying effects of feature values—an area where SHAP excels. SHAP not only highlights key features but also illustrates how their values, whether high or low, contribute to the prediction. This multi-faceted approach offers a more nuanced understanding of feature importance, making it more suitable for clinical applications where understanding interactions between symptoms and biomarkers is essential.

3.2.2 *Local interpretability with SHAP and LIME*

For local interpretability, we applied both SHAP and LIME to explain feature contributions for individual instances. Figure 3 shows the SHAP decision plot, providing a detailed breakdown of how individual features cumulatively deviated the output of the model from the baseline. For a given instance, Serum Creatinine and GFR were the most significant contributors pushing the prediction towards CKD. The interactions between features and their combined impact on the model's decision-making process are depicted in this image. In contrast, Figure 4 illustrates the LIME explanation for the same instance. LIME provides a more localized and simplified interpretation, identifying GFR and Serum Creatinine as key factors influencing the prediction.

It highlights specific threshold values, such as GFR $\geq$ 93.31, which strongly push the model towards predicting CKD. While LIME offers a clear, instance-specific explanation, it assumes feature independence, potentially oversimplifying the underlying interactions.

Both SHAP and LIME identified Serum Creatinine and GFR as the top contributors to CKD prediction.

However, SHAP provides a global perspective, offering insight into feature interactions and cumulative contributions across the model, whereas LIME delivers a more straightforward local explanation, suitable for understanding decisions on a per-instance basis. SHAP provides a comprehensive global view, while LIME excels in localized, instance-specific explanations. Feature Interactions: SHAP captures feature interactions, while LIME assumes feature independence. SHAP offers a more nuanced and detailed understanding, while LIME provides clear binary influences, ideal for interpreting immediate decision boundaries. Together, these complementary methods provide valuable insights into the model's interpretability, ensuring transparency in both global feature importance and local predictions, essential for clinical decision-making in CKD.

4 CONCLUSION

In order to improve model interpretability, we used Explainable AI (XAI) methodologies in this study and assessed machine learning classifiers for the prediction of Chronic Kidney Disease. The model with the highest performance and accuracy was XGBoost (93.49%) and robust precision and recall. By integrating SHAP, LIME, and Permutation Feature Importance, we provided both global and local insights into feature contributions. Serum Creatinine and Glomerular Filtration Rate (GFR) were consistently identified as the most critical predictors across all XAI methods, affirming their clinical significance. SHAP's ability to reveal feature interactions and LIME's instance-specific clarity complement each other, offering a comprehensive view of model behaviour. This research embarks the potential of incorporating machine learning models with explainability techniques to foster trust in AI-driven healthcare tools, ensuring more transparent and informed clinical decision-making for CKD diagnosis and

management. Ultimately, this approach fosters the ethical integration of explainable AI in Healthcare which boosts the diagnostic accuracy and clinician confidence. Future work will focus on deploying the proposed model in clinical environments and integrating it with electronic health record (EHR) systems to facilitate real-time decision support.

REFERENCES

[1] Wouters, O.J., O'donoghue, D.J., Ritchie, J., Kanavos, P.G., and Narva, A.S. (2015). Early chronic kidney disease: diagnosis, management and models of care. *Nature Reviews Nephrology*, 11 (8), 491–502.

[2] Baumgarten, M., and Gehr, T. (2011). Chronic kidney disease: detection and evaluation. *American Family Physician*, 84(10), 1138–1148.

[3] Bikbov, B., Purcell, C.A., Levey, A.S., Smith, M., Abdoli, A., Abebe, M., ... and Owolabi, M.O. (2020). Global, regional, and national burden of chronic kidney disease, 1990–2017: a systematic analysis for the Global Burden of Disease Study 2017. *The Lancet*, 395(10225), 709–733.

[4] Song, J., Pan, T., Xu, Z., Yi, W., Pan, R., Cheng, J., ... and Su, H. (2024). Asystematic analysis of chronic kidney disease burden attributable to lead exposure based on the global burden of disease study 2019. *Science of the Total Environment*, 908, 168189.

[5] Islam, R., Sultana, A., and Islam, M.R. (2024). A comprehensive review for chronic disease prediction using machine learning algorithms. *Journal of Electrical Systems and Information Technology*, 11(1), 27.

[6] Rahman, M.M., Al-Amin, M., and Hossain, J. (2024). Machine learning models for chronic kidney disease diagnosis and prediction. *Biomedical Signal Processing and Control*, 87, 105368.

[7] Laghmati, S., Hamida, S., Hicham, K., Cherradi, B., and Tmiri, A. (2024). An improved breast cancer disease prediction system using ML and PCA. *Multimedia Tools and Applications*, 83(11), 33785–33821.

[8] Naser, M.A., Majeed, A.A., Alsabah, M., Al-Shaikhli, T.R., and Kaky, K.M. (2024). A review of machine learning's role in cardiovascular disease prediction: recent advances and future challenges. *Algorithms*, 17(2), 78.

[9] Mitra, U., and Rehman, S.U. (2025). Leveraging AI and machine learning for next-generation clinical decision support systems (CDSS). In *AI-Driven Innovation in Healthcare Data Analytics* (pp. 83–112). IGI Global Scientific Publishing.

[10] Das, A., Choudhury, D., and Sen, A. (2024). A collaborative empirical analysis on machine learning based disease prediction in health care system. *International Journal of Information Technology*, 16(1), 261–270.

[11] Hassija, V., Chamola, V., Mahapatra, A., Singal, A., Goel, D., Huang, K., ... and Hussain, A. (2024). Interpreting black-box models: a review on explainable artificial intelligence. *Cognitive Computation*, 16(1), 45–74.

[12] Doshi-Velez, F., and Kim, B. (2017). Towards a rigorous science of interpretable machine learning. *arXiv preprint arXiv*:1702.08608.

[13] Ribeiro, M.T., Singh, S., and Guestrin, C. (2016). "Why should I trust you?": explaining the predictions of any classifier. *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, 1135–1144.

[14] Vimbi, V., Shaffi, N., and Mahmud, M. (2024). Interpreting artificial intelligence models: a systematic review on the application of LIME and SHAP in Alzheimer's disease detection. *Brain Informatics*, 11(1), 10.

[15] Ahmed, S., Kaiser, M.S., Hossain, M.S., and Andersson, K. (2024). A comparative analysis of LIME and SHAP interpreters with explainable ML-based diabetes predictions. *IEEE Access.*